

Corrosion Severity Grading on Industrial Machinery: Zero-/Few-Shot Vision-Language Models vs. Supervised CNN Baselines

Industry Collaboration with Everllence SE (formerly MAN Energy Solutions)

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Abstract

Corrosion is a dominant degradation mechanism for the marine and stationary machinery Everllence services through its PrimeServ network [13]. Supervised CNN-based corrosion detection is well established [3, 1], but it demands abundant labels per component and coating, and it answers only “is there corrosion?”. The economically relevant task, *severity grading* against standard scales (ASTM D610 [5]), suffers from fuzzy grade boundaries and scarce labels [4]. This thesis asks one focused question: *can vision-language models (VLMs), operating zero-/few-shot with natural-language justifications, grade severity competitively with supervised CNNs trained on the same data?* On public datasets with severity annotations [2, 12], a fine-tuned CNN (with a label-efficiency curve at 100%/50%/10% of labels) is compared against Qwen2.5-VL [9] zero-/few-shot and a CLIP reference [6], using ordinal metrics and a boundary confusion analysis, plus an expert review of VLM justifications by Everllence surveyors.

Motivation and Gap

Everllence assets operate in salt-laden, thermally cycled environments where corrosion drives maintenance scheduling; inspection imagery accumulates in service reports but assessment is manual and subjective. Supervised CNNs work [1, 3] yet each component geometry effectively demands new annotations — the adoption blocker for a service organisation with heterogeneous, confidential imagery. VLMs offer a different cost profile: zero-/few-shot operation and justification text. Zero-shot industrial inspection is an established research line [7, 8], and recent surveys flag reasoning-oriented corrosion assessment as open territory [11]: existing foundation-model work on corrosion is segmentation-oriented [10]. Whether a VLM can grade *severity* competitively with a CNN on equal data has not been measured; that single measurement — with a label-efficiency analysis quantifying when the zero-label regime wins — is the thesis.

Feasibility

- The supervised half is a validated, bounded workload: applied studies of the same shape (YOLOv8 vs. Detec-tron2 on ship-hull plates) are published at similar scope [1].
- All quantitative work on public data: bridge corrosion with condition ratings [2], ship-hull dry-dock imagery [12]; a small non-confidential Everllence sample serves a qualitative case study only.
- Methods are mature: CNN fine-tuning is routine; Qwen2.5-VL is open-weight with documented inference tooling [9]; no VLM fine-tuning in core scope (LoRA optional). Single-GPU + Azure compute suffices.

Research Questions

1. **RQ1:** How close does a zero-/few-shot VLM come to a supervised CNN on a common four-grade severity scale — and how does the gap change as CNN training labels shrink (label efficiency)?
2. **RQ2:** Within a fixed prompt budget, how much do few-shot exemplars and region cropping improve VLM grading accuracy and run-to-run stability?
3. **RQ3:** Are VLM errors distributed differently at fuzzy grade boundaries (ordinal distance of mistakes), and do Everllence surveyors judge the VLM’s justifications plausible?

Methodology

- **Data:** bridge corrosion dataset with native condition ratings [2] as primary benchmark; marine sets [12] as transfer/case-study material. Labels harmonised once onto an ASTM D610-aligned four-grade scale (documented mapping); source-wise splits.
- **Models:** fine-tuned ResNet/EfficientNet classifier at 100%/50%/10% labels; CLIP zero-shot reference [6]; Qwen2.5-VL zero-shot and $k \in \{2, 4\}$ few-shot with two prompt variants (full image vs. crop); fixed decoding, three repeated runs.
- **Evaluation:** grade MAE, quadratic weighted kappa, macro-F1; boundary confusion analysis; VLM output stability; expert review of a stratified justification sample by 2–3 surveyors on a fixed rubric.
- **Scope/risk:** segmentation research, corrosion-type taxonomy, deployment out of scope. If label harmonisation



across datasets proves noisy, the bridge dataset alone carries RQ1–RQ3. A negative VLM result is decision-relevant: it justifies the annotation investment it would otherwise replace.

Contribution and Value for Everllence

(1) The first direct equal-data comparison of zero-/few-shot VLM severity grading vs. supervised CNNs, with a label-efficiency analysis; (2) a documented ASTM-aligned cross-dataset severity protocol with ordinal metrics; (3) an expert-reviewed assessment of VLM justification quality, plus released code and prompts. For Everllence: an annotation-investment decision — if few-shot VLM grading approaches CNN accuracy, costly component-specific labelling campaigns can be reduced; if not, the measured gap justifies them.

Timeline

Month	Work
1	Literature review; data acquisition; severity harmonisation and splits; grading rubric.
2	CNN baseline: fine-tuning + label-efficiency runs; CLIP reference.
3	VLM pipeline; zero-shot evaluation; prompt variants.
4	Few-shot experiments; stability runs; full RQ1/RQ2 benchmark.
5	Boundary/error analysis (RQ3); expert review; Everllence case study; optional LoRA extension.
6	Thesis writing, revision, code/prompt release, submission.

References

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